IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MATHEMATICS 1B

Tuesday, 28 April 2026, 2:30 PM

Time allowed: 2:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks (25% each).

NO CALCULATORS ALLOWED. Formula sheet will be provided.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : D. Nucinkis
Second Marker(s) : T. Pike

© Imperial College London

THE QUESTIONS

1. a) Two matrices are given as

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 3 \\ 1 & \alpha & 1 \end{pmatrix}, \quad \text{and} \quad B = \begin{pmatrix} 3 & 2 & 4 \\ 3 & 2 & -1 \\ 1 & 1 & -1 \end{pmatrix}$$

(i) Let $\alpha = 1$. Find $\det(A)$, as well as $\det(B)$ and $\det(AB)$. Finally, use Gauss-Jordan elimination to obtain A^{-1}. [6]

(ii) Let $\alpha = 3$. Give conditions on $\underline{\mathbf{b}} = (2, \beta, 1)^T$ so that the matrix equation $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$ has no solutions, or infinitely many solutions. Can the system have a unique solution? Either way, explain why. Finally, for the case of infinitely many solutions, find these and give a geometric interpretation. [6]

b) (i) Use the definition of the Laplace transform to prove the first shift theorem:

$$\mathscr{L}[e^{at} f(t)] = \bar{f}(s-a) ,$$

where $\mathscr{L}[f(t)] = \bar{f}(s)$. State any necessary condition. [4]

(ii) Use the first shift theorem to obtain the inverse Laplace transform of

$$\frac{5}{(s+3)(s^2+4s+13)} .$$

[5]

(iii) Use Laplace transforms and (ii) to solve the Ordinary Differential Equation

$$\frac{d^2x}{dt^2} + 4\frac{dx}{dt} + 13x = 5e^{-3t}$$

which satisfies the initial conditions $x(0) = 1$ and $x'(0) = -2$. [4]

2. a) Obtain $F(\omega)$, the Fourier transform of a real signal $f(t)$, defined as

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \,, \\ e^{-t}\cos(t) & \text{for } t \geq 0 \,. \end{cases}$$

Show that we can write

$$F(\omega) = \frac{1+i\omega}{(1+i\omega)^2+k} \,,$$

and provide the value of the real number k. [7]

b) A function of two variables is given as

$$f(x,y) = x\cos y + y\sin x \,.$$

(i) Obtain the Taylor series expansion of $f(x,y)$, around the point $(\pi/2, 0)$, to second order. [5]

(ii) Hence find the equation of the tangent plane to the surface $z = f(x,y)$, at the point $(\pi/2, 0)$. [4]

c) (i) Use the method of Lagrange multipliers to find the point on the plane with cartesian equation

$$5x + y - 4z = 11$$

which is nearest to the point $(3, 1, -4)$. [5]

[*Hint: a distance function and its square have the same minimum.*]

(ii) Confirm the result from (i) by finding the intersection of the plane and a line with vector equation $\underline{\mathbf{r}} = \underline{\mathbf{a}} + \mu\underline{\mathbf{b}}$, where μ is any real number, and $\underline{\mathbf{a}}$ and $\underline{\mathbf{b}}$ need to be found. [4]

3. a) Three vectors in $\mathbb{R}^3$ are given as $\underline{\mathbf{x}} = (1,0,3)^T$, $\underline{\mathbf{y}} = (2,1,4)^T$, and $\underline{\mathbf{z}} = (-1,1,0)^T$, the position vectors of points X, Y and Z. A line L goes through X and Y. Let the line have direction $\underline{\mathbf{a}}$. Find the projection of the vector $\overrightarrow{XZ}$ onto $\underline{\mathbf{a}}$. Hence, or otherwise, find the point Q on line L which is nearest to Z, and find the shortest distance from Z to L. Explain your reasoning and provide a sketch. [7]

b) A double integral is given by

$$I = \int\int_R (x+y)^n \, f(x^2-y^2) \, dx \, dy \, ,$$

where R is the region of the $xy-$plane enclosed by the coordinate axes and the line $y = 1-x$.

(i) Use the change of variables $u = x^2 - y^2$, $v = x+y$ to rewrite the integral as

$$I = \frac{1}{2} \int\int_{R*} v^{n-1} f(u) \, du \, dv \, ,$$

and sketch the region R^* in the $uv-$plane.

(ii) Rewrite the double integral in two ways: (1) integrating first with respect to u, and (2) integrating first with respect to v.

(iii) Finally, evaluate the integral for the case $n = 2$ and $f(u) = e^u$. You can choose to integrate first with respect to u or v, but one of these will be easier. [10]

c) If A is a 4×3 matrix, what can you say about rank, pivot and free columns, nullspace and column space of A? In each case, what does this imply for the solutions of $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$? Assume $\underline{\mathbf{b}}$ is not the zero vector. When solutions exist, give a geometric description. [8]

4. a) Obtain constants α and β, so the first-order Ordinary Differential Equation (ODE)

$$\alpha xt + \cos x \cos t \; + \frac{1}{t} \; + \left(t^2 - x^2 + \beta \sin x \sin t \; \right) \frac{dx}{dt} = 0$$

is exact. Solve the ODE, subject to the condition $x = 0$ when $t = \pi$. [7]

b) Use a Fourier Series to show that, for $0 < x < \pi$,

$$\sin x = \frac{2}{\pi} - \frac{4}{\pi}\left(\frac{1}{1\cdot 3}\cos(2x) + \frac{1}{3\cdot 5}\cos(4x) + \frac{1}{5\cdot 7}\cos(6x) + \frac{1}{7\cdot 9}\cos(8x) + \cdots\right) . \text{ [10]}$$

You may assume the identity $\sin[(m-n)x] + \sin[(m+n)x] = 2\sin(mx)\cos(nx)$.

c) Let a matrix be given as

$$A \; = \; \begin{pmatrix} -1 & 2 & 1 & 0 \\ 0 & 1 & -2 & 3 \\ 2 & -3 & -4 & 3 \\ 0 & -2 & 4 & -6 \end{pmatrix}$$

Use Gaussian elimination to obtain the echelon form of A, identify rank, pivot and free columns. Hence find the nullspace $N(A)$ and column space $C(A)$ as the linear combinations of independent vectors. Finally, obtain a particular solution of $A\underline{\mathbf{x}} = \underline{\mathbf{b}} = (3, -4, -10, 8)^T$, and with no further calculation, write down the general solution. [8]

MATHEMATICAL FORMULAE

1 Vector Algebra

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \;=\; (a_1, a_2, a_3)$$

Scalar(Dot) Product $\quad \mathbf{a}\cdot\mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$

Vector (Cross) Product

$$\mathbf{a}\times\mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\mathbf{i} + (a_3b_1 - a_1b_3)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k}.$$

Triple vector product $\quad \mathbf{a}\times(\mathbf{b}\times\mathbf{c}) = (\mathbf{a}\cdot\mathbf{c})\mathbf{b} - (\mathbf{a}\cdot\mathbf{b})\mathbf{c}$

Triple scalar product

$$\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}) = \mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2 - a_1b_3c_2 - a_2b_1c_3 - a_3b_2c_1$$

2 Series

$$(1+x)^n = 1+nx+\frac{n(n-1)}{2!}x^2+\frac{n(n-1)(n-2)}{3!}x^3+\ldots+\frac{n(n-1)(n-2)\cdots(n-r+1)}{r!}x^r+\ldots \;(n\,\text{arbitrary}\,,|x|<1)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \ldots + \frac{x^r}{r!} + \ldots \qquad \text{for all } x$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots + (-1)^{r+1}\frac{x^r}{r} + \ldots \qquad \text{for } -1 < x \leq 1$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots + \frac{(-1)^r x^{2r}}{(2r)!} + \ldots \qquad \text{for all } x$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots + \frac{(-1)^r x^{2r+1}}{(2r+1)!} + \ldots \qquad \text{for all } x$$

3 Trigonometric Identities and Hyperbolic Functions

$$\sin(A+B) = \sin A\cos B + \sin B\cos A\,, \qquad \cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(iz) = \cosh z\,, \quad \cosh(iz) = \cos z\,, \quad \sin(iz) = i\,\sinh(z)\,, \quad \sinh(iz) = i\,\sin z$$

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\cos(x)\cos(y) = \frac{1}{2}\cos(x-y) + \frac{1}{2}\cos(x+y)$$

$$a\cos x + b\sin(x) = c\cos(x+\theta) \quad \text{where} \quad c = \sqrt{a^2+b^2} \quad \text{and} \quad \theta = \tan^{-1}\left(-\frac{b}{a}\right)$$

4 Differential Calculus

1. Leibnitz' Formula: $D^n(fg) =$

$$D^n f\, g + \binom{n}{1} D^{n-1} f\, Dg + \binom{n}{2} D^{n-2} f\, D^2 g + \cdots + \binom{n}{r} D^{n-r} f\, D^r g + \cdots + f D^n g$$

where $\begin{pmatrix} n \\ r \end{pmatrix} = \dfrac{n(n-1)(n-2)\cdots(n-r+1)}{r!}$.

2. Taylor expansion of $f(x)$ about the point $x = a$

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

where

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \qquad a < c < x$$

Putting $x = a + h$ gives the form

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!} f''(a) + \frac{h^3}{3!} f'''(a) + \ldots$$

3. Taylor expansion of $f(x, y)$ about the point (a, b):

$$f(x,y) = f(a,b) + \frac{1}{1!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right] f(a,b) + \frac{1}{2!}\left[(x-a)\frac{\partial}{\partial x} + (y-b)\frac{\partial}{\partial y}\right]^2 f(a,b) + \cdots$$

4. Chain rule for partial derivatives: $f = f(x, y)$

(i) If $x = x(t)$ and $y = y(t)$ then $f = F(t)$ and

$$\frac{dF}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}$$

(ii) If x and y are expressed in terms of new variables u and v by $x(u, v)$ and $y(u, v)$ then $f = F(u, v)$ and

$$\frac{\partial F}{\partial u} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial u}, \quad \text{and} \quad \frac{\partial F}{\partial v} = \frac{\partial f}{\partial x}\,\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\,\frac{\partial y}{\partial v}$$

(iii) If $y = y(x)$ then $f = F(x)$ and

$$\frac{dF}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{dy}{dx}$$

5. Stationary points of $f(x, y)$ occur where $f_x = 0$ and $f_y = 0$. Let (a, b) be a stationary point, and consider $D = f_{xx}(a,b) f_{yy}(a,b) - [f_{xy}(a,b)]^2$. Then

(a) If $D > 0$ and $f_{xx}(a, b) < 0$, then (a, b) is a maximum;

(b) If $D > 0$ and $f_{xx}(a, b) > 0$, then (a, b) is a minimum;

(c) If $D < 0$ then (a, b) is a saddle point;

(d) if $D = 0$, more information is required.

6. Differential Equations

(i) The first order linear equation $\dfrac{dy}{dx} + p(x)y = r(x)$ has an integrating factor $I(x) = \exp\left\{\displaystyle\int p(x)\, dx\right\}$ which allows one to write the DE in the form $\dfrac{d}{dx}(I(x)y) = I(x)r(x)$.

(ii) The first-order equation $P(x, y)\dfrac{dy}{dx} + Q(x, y) = 0$ is exact if $\dfrac{\partial P}{\partial x} = \dfrac{\partial Q}{\partial y}$.

5 Integral Calculus

1. An important substitution: $\tan(\theta/2) = t$; then

$$\sin\theta = \frac{2t}{1+t^2}\,, \quad \cos\theta = \frac{1-t^2}{1+t^2}\,, \quad d\theta = \frac{2\,dt}{1+t^2}\,.$$

2.
$$\int \frac{dx}{(a^2-x^2)^{1/2}} \;=\; \sin^{-1}\left(\frac{x}{a}\right)\,, \quad |x| < a\,.$$

3.
$$\int \frac{dx}{(a^2+x^2)^{1/2}} \;=\; \sinh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}+1\right)^{1/2}\right]\,.$$

4.
$$\int \frac{dx}{(x^2-a^2)^{1/2}} \;=\; \cosh^{-1}\left(\frac{x}{a}\right) \;=\; \ln\left[\frac{x}{a} + \left(\frac{x^2}{a^2}-1\right)^{1/2}\right]\,.$$

5.
$$\int \frac{dx}{a^2+x^2} \;=\; \left(\frac{1}{a}\right)\tan^{-1}\left(\frac{x}{a}\right)\,.$$

6 Fourier Series

The following formulae assume that $f(x)$ satisfies the Dirichlet conditions and is periodic with period T, i.e. $f(x+T) = f(x)$. The general Fourier Series for $f(x)$ is

$$f(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty}\left[a_n\cos\left(\frac{2n\pi x}{T}\right) + b_n\sin\left(\frac{2n\pi x}{T}\right)\right]$$

where

$$a_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\;\cos\left(\frac{2n\pi x}{T}\right)\,dx\,,\; b_n = \frac{2}{T}\int_{-T/2}^{T/2} f(x)\;\sin\left(\frac{2n\pi x}{T}\right)\,dx\,,\; n = 0,1,2,3,\ldots$$

The series converges to $f(x)$ at points of continuity and to the mean value $\frac{1}{2}(f(x_+) + f(x_-))$ at points where $f(x)$ is discontinuous.

The complex form of the Fourier series is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/T}\,, \text{ where } c_n = \frac{1}{T}\int_{-T/2}^{T/2} f(x)e^{-i2n\pi x/T}\,dx \text{ for every integer } n.$$

Parseval's Theorem: If the complex Fourier series of the T-periodic function $f(x)$ has coefficients c_n, and the real Fourier series of $f(x)$ has coefficients a_n and b_n, then

$$\frac{1}{T}\int_{-T/2}^{T/2} f^2(x)\,dx \;=\; \sum_{n=-\infty}^{\infty}|c_n|^2\,. \qquad \text{and} \qquad \frac{2}{T}\int_{-T/2}^{T/2} f^2(x)\,dx = \frac{1}{2}a_0^2 + \sum_{n=1}^{\infty} a_n^2 + b_n^2\,.$$

7 Fourier Transforms

The following formulae assume that $g(t)$ and $f(t)$ are the sum of one or more components that obey the Dirichlet conditions. The Fourier transform pair is

$$G(\omega) = \int_{-\infty}^{\infty} g(t) e^{-i\omega t}\, dt\,, \quad \Longleftrightarrow \quad g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{i\omega t}\, d\omega\,,$$

where $G(\omega) = \mathcal{F}\left[g(t)\right]$ is the *Fourier transform* of $g(t)$, while $g(t) = \mathcal{F}^{-1}\left[G(\omega)\right]$ is the *inverse Fourier transform* of $G(\omega)$.

The Dirac delta function $\delta(t)$ satisfies $\displaystyle\int_{-\infty}^{\infty} \delta(t)dt = 1, \qquad \int_{-\infty}^{\infty} f(t)\delta(t-a)dt = f(a)$

For a periodic function $f(t)$ with angular frequency ω_0, and complex Fourier series as given,

$$f(t) = \sum_{n=-\infty}^{\infty} F_n e^{in\omega_0 t} \qquad \Longrightarrow \qquad \mathcal{F}\left[f(t)\right] = F(\omega) = 2\pi \sum_{n=-\infty}^{\infty} F_n \delta(\omega - n\omega_0)\,.$$

Time/Frequency-shift: If $\mathcal{F}\left[f(t)\right] = F(\omega)$ then $\mathcal{F}[f(t-t_0)] = e^{-i\omega t_0} F(\omega)$ and $\mathcal{F}\left[e^{i\omega_0 t} f(t)\right] = F(\omega - \omega_0)\,.$

Symmetry/Duality property: $\mathcal{F}\left[g(t)\right] = G(\omega) \Longleftrightarrow \mathcal{F}\left[G(t)\right] = 2\pi g(-\omega)$

$\mathcal{F}\left[\cos(\omega_0 t)\right] = \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

8 Laplace Transforms

$f(t)$	$\mathcal{L}[f(t)] = \bar{f}(s)$
1	$\dfrac{1}{s}\,, \quad \mathrm{Re}(s) > 0$
$t^n\,, \quad n = 1, 2, 3, \ldots$	$\dfrac{n!}{s^{n+1}}\,, \quad \mathrm{Re}(s) > 0$
$\sin(at)$	$\dfrac{a}{s^2 + a^2}\,, \quad \mathrm{Re}(s) > 0$
$e^{at} f(t)$	$\bar{f}(s - a)$
$\dfrac{df}{dt}$	$s\bar{f}(s) - f(0)$
$\delta(t - a)$	$e^{-as}\,, \quad a > 0$

$f(t)$	$\mathcal{L}[f(t)] = \bar{f}(s)$
e^{at}	$\dfrac{1}{s - a}\,, \quad \mathrm{Re}(s) > a$
$H(t - a)$	$\dfrac{e^{-as}}{s}\,, \quad \mathrm{Re}(s) > 0$
$\cos(at)$	$\dfrac{s}{s^2 + a^2}\,, \quad \mathrm{Re}(s) > 0$
$f(t - a)H(t - a)$	$e^{-as}\bar{f}(s)$
$\dfrac{d^2 f}{dt^2}$	$s^2\bar{f}(s) - sf(0) - \dfrac{df}{dt}(0)$
$\displaystyle\int_0^t f(u)g(t-u)\, du$	$\bar{f}(s)\bar{g}(s)$